Bryan Wilcutt – Embedded Systems Engineer

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ENGINEERING SUMMARY

Principal Embedded Systems Engineer and Engineering Manager specializing in real-time Linux, secure communication systems, and mission-critical aerospace/defense platforms. 20+ years leading development of embedded architectures, device drivers, BSPs, and RF/BLE systems across PIC32, ARM, Nordic, and custom hardware platforms. Known for solving deep hardware/software integration issues, accelerating bring-up, and delivering robust, certifiable embedded systems.

Seeking a full-time, long-term position.

CORE TECHNICAL SKILLS

• Embedded C/C++ • Python • RTOS • Bare-Metal Development

• Linux Kernel • Device Drivers • Board Bring-Up • BSP Development

• MCUs: PIC32, ARM AM437x, Nordic (nRF52), Renesas, Blackfin DSP, Intel, Motorola

• Interfaces: I2C, SPI, UART, CAN, ECAN, RS485, BLE, Cellular, RF

• Tools: GCC, LLVM, GDB, JTAG, Oscilloscopes, Logic Analyzers

• Protocols: BLE, RF, Satellite Comms, Secure Embedded Protocols, Telephony Call Control, Others

• Build Systems: Git, GitLab CI/CD, Conan, Jfrog, Cmake, Others

FULL ENGINEERING EXPERIENCE

Lead Embedded Engineer – AIIR Products (Startup Project Role - 2023–2025)

- Assisted startup company in the design and development of an AI Based HVAC system.
- Developed embedded software for ARM AM437X Linux home automation hub.
- Implemented device drivers, sensor interfaces, and hardware test systems.
- Designed Redis-based embedded data system and automated CI/CD.
- Developed software to track customer warranties, print labels, and tie everything together with Redis.
- Performed system-level debugging across hardware, kernel, and userland.
- Developed all testing applications used by factory.

Lead Embedded Software Engineer – Elbit (Full Time - 2022–2023)

- Developed PIC32 embedded firmware for F-35 EO/IR camera systems. Fixed numerous bugs.
- Built targeting algorithms for helmet-mounted avionics.
- Integrated FPGA interfaces and real-time imaging control paths.
- Developed company's custom F-35 radio test software suite in Microsoft Visual Studio.

Lead Embedded Engineer – SmartStart (Project Role - 2021–2022)

- Assisted startup company in Proof of Concept of BLE communication systems.
- Developed BLE-enabled Nordic-based automotive relay systems in C.
- Built embedded stack enabling vehicle interlock communication.
- Designed relay-gang automated test platform (Nordic platform).
- BLE-to-Starter-Relay was unprovable due to vehicle interference.

Lead Embedded Engineer – Apptricity (Project Role - 2020–2021)

- Startup Company in the development of RF Tag (passive/active) communications.
- Developed embedded Linux + ThreadX IoT tracking systems.
- Implemented UART, BLE, WIFI, Cellular, and RF drivers.
- Integrated Impinj, Ublox, Zentri wireless modules into product line.
- Worked with software/hardware teams to achieve a record breaking 20-mile range for BLE Beacon product.

Embedded Software Engineering Manager – IDEX (Management Role - 2018–2019)

- Led embedded development of fuel metering and mass-sensor products.
- C/C++ development on ARM and Cortex-based embedded systems.
- Supported hardware bring-up and system validation.
- Increased bootup time by 20%, fixed several hardware issues, improved certification process.
- Managed 9 cross functional team of engineers in Chicago, Los Angeles, and Tampa Bay for LCR.IQ product.

Lead Embedded Engineer – Leonardo/Selex (Project Role - 2017–2018)

- Developed avionics embedded systems (ILS, VOR, DME).
- Ported 200KLOC Linux systems to modern GCC/LLVM toolchains.
- Implemented SELinux hardening for aerospace-certified products.
- Belgium: Assisted EU in VOR/DME testing at Belgium Airport.

Lead Software Engineer/Product Manager – LynxSpring (Management Role - 2015–2017)

- Assisted and led startup company in the design and development of Building Automation and Control products.
- Developed Linux BeagleBone environment and custom device drivers.
- Implemented AM335x PRU-based embedded control modules.
- Built BACnet, Modbus, CAN, and sensor interfaces.
- Worked with customers in improving Onyxx product line.

Software Team Lead – JCM Global (Management Role - 2011–2015)

- Developed C/C++ embedded firmware for global bill-validation systems.
- Performed hardware-level debugging using oscilloscopes/JTAG.
- Led neural-algorithm integration for banknote recognition.
- Managed team of 8 engineers for software development of AI image analysis system.
- Managed team of 22 engineers for bill validator validation.
- Managed team of 3 engineers for notes collection world-wide.
- Bill Equipment Manufacturer representative to Federal Reserve.
- Team lead of Patent Development Group.
- Belgium: Assisted EU in redesign of EU currency (ink).
- Japan: Development of "long edge" bill validator.
- Developed company's currency data storage system.
- Developed "ICB 2.0" (Intelligent Cash Box) using NFC communication.
- Developed 25+ patents.

Software Engineer – Apriva ISS (Project Role - 2010–2011)

- Perform software design and development for an NSA communications contract.
- Architected AprivaTalk secure embedded protocol (RSA, AES, PKI).
- Built cross-platform embedded communication system in C/C++.
- Developed phone based application to replace SMEPED devices.
- Project was completed and transferred to the NSA.

Early Career (Full Time Roles - 1996–2009) — Comtech EF Data, Motorola, Orbital Sciences

- Developed embedded C/C++ systems for satellite communications (Iridium), missile flight controllers (Orbital), and defense applications. Led small teams and contributed to compiler/toolchain evaluation, device drivers, applications, and protocol design.

EDUCATION

Doctorate of Computer Science (ABD) – Institute for Advanced Studies, 2012

M.S. Management Science – Colorado Technical University, 2007

B.S. Computer Science – Hawaii Pacific University, 1993
Six Sigma Green Belt – Simplilearn
Project Management Professional – PMI

OTHER

14 U.S. Patents (Embedded systems, RF, secure communications). List available upon request.
Awards: Diebold 2012 Manufacturer's Award (JCM), International Fueler's Award 2019 (IDEX), Developer's Award (1998, 2000 - Motorola), 2017 Frost & Sullivan North American Award for Product Design (LynxSpring)
Authored Satellite NMCS Protocol (IANA Port 7023 UDP/TCP)